YouTube Decoded

How YouTube Knows What You Want to Watch

Foundations in AI & Machine Learning | Session 2

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Foundations in AI & Machine Learning | Session 2

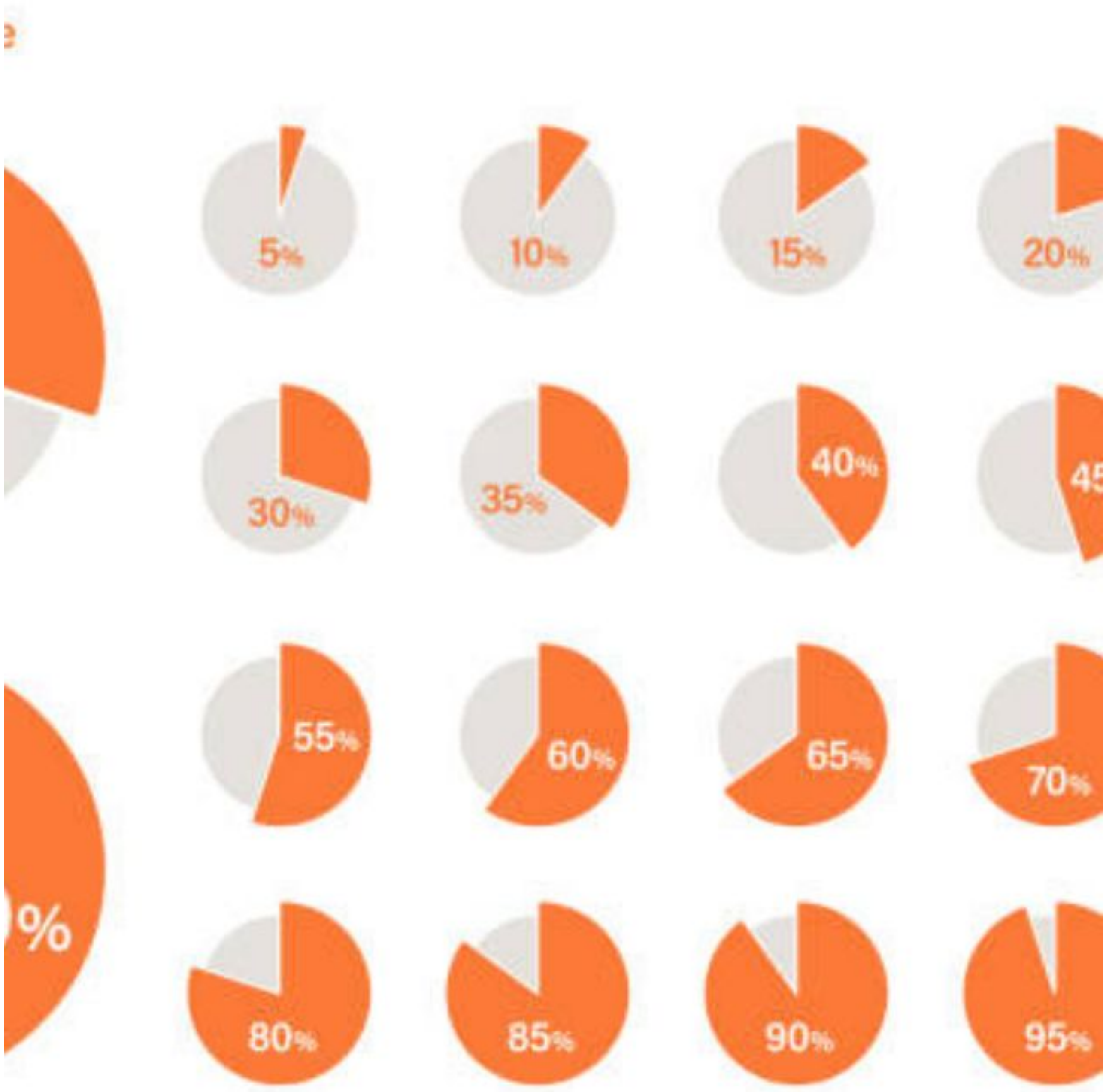
YouTube knows what you want to watch...

...before you do.

70%

of all YouTube watch time comes from recommendations

Source: YouTube Internal Data, 2024



The Impossible Problem

313,000

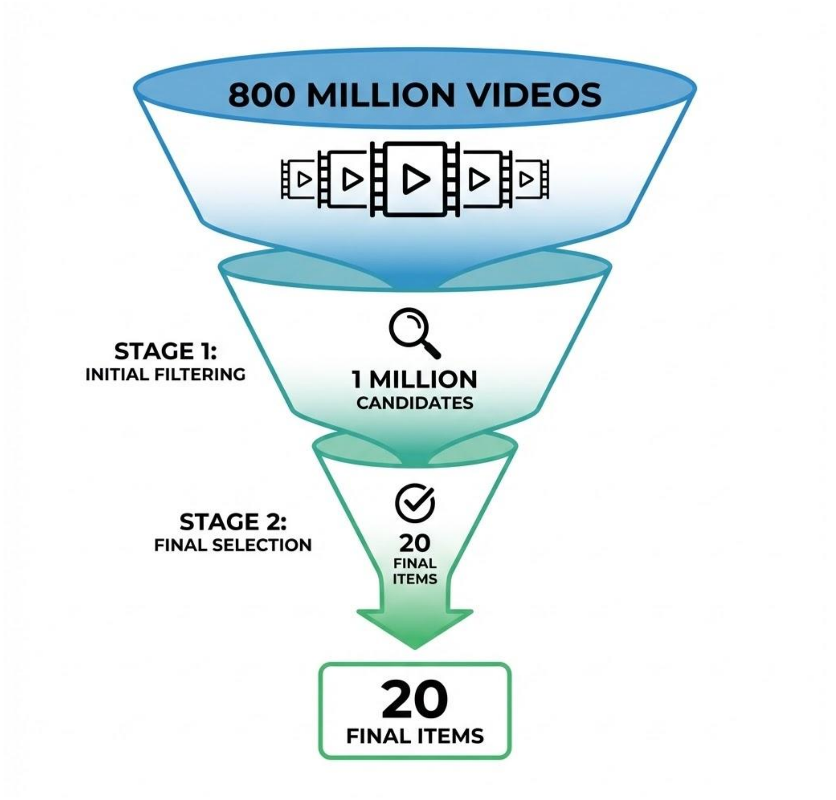
Restaurants (Zomato)

VS

800,000,000

Videos (YouTube)

Filters down to just 20 videos



Zomato vs YouTube Scale

① Filtering → 313000 → 20
② Cold Start →
③ Feedback loops →

Dimension	Zomato	YouTube
Total Options	313,000	800,000,000
Shown to User	~20	~20
Decision Time	< 500ms	< 500ms
Revenue (2024)	$1 Billion	$54.2 Billion

122,000,000

2.7B

8 Billion

2.7 B

Monthly Active Users

More than 1/3 of all humans on Earth

114,000 Years

1 Billion

Hours of video watched EVERY DAY

That's 114,000 years of content consumed daily

3,00,000

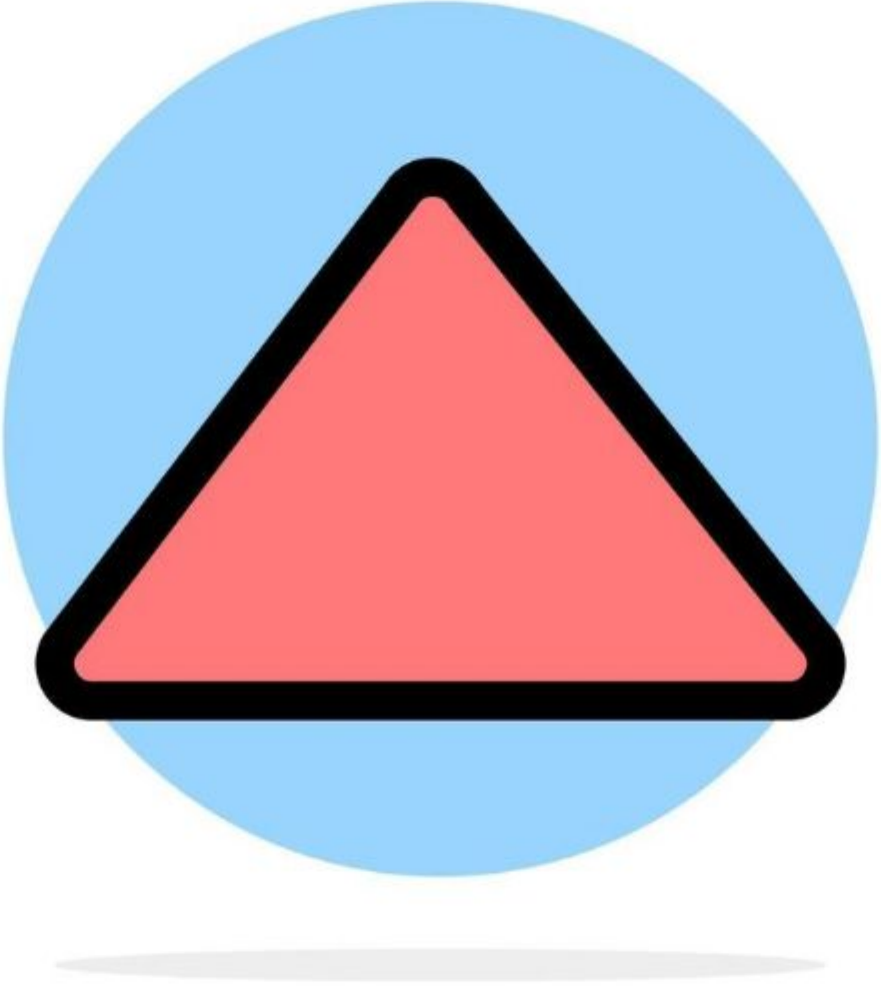
7,20,000 hou

500 Hours

of new video uploaded every

MINUTE

720,000 hours of new content daily



2.7 Billion x 12

$54.2B

Total Revenue in 2024

More than Netflix's entire business

$36.1B ads + $14.5B subscriptions

~ $ 20

Why Humans Can't Solve This

Time

800M videos × 1s = 25 YEARS to scan

Scale

2.7B users need recommendations

Speed

Response needed in < 500ms

Imagine You're a Librarian...

- Your library is the size of a COUNTRY
- You have 800 MILLION books
- 122 MILLION people walk in EVERY DAY
- Each needs 20 perfect books in HALF A SECOND
- Oh, and 500 new books arrive every MINUTE

2006 Youtube → Google

2010 ClickBait

2012 → Clicks

Watch time

What YouTube Tracks

The Data Behind the Magic

Why Clicks Don't Matter Anymore

A

Clickbait Video

1M clicks

15 second watch time

Result: User Hated It

1M Click

250,000 Mins

B

Quality Video

100K clicks

8 minute watch time

Result: User Loved It

800,000

YouTube's Revolutionary Change

Success = Clicks

Success = Watch Time

= Satisfaction

This single change killed clickbait culture.

What You SAY vs What You DO

Explicit (10%)

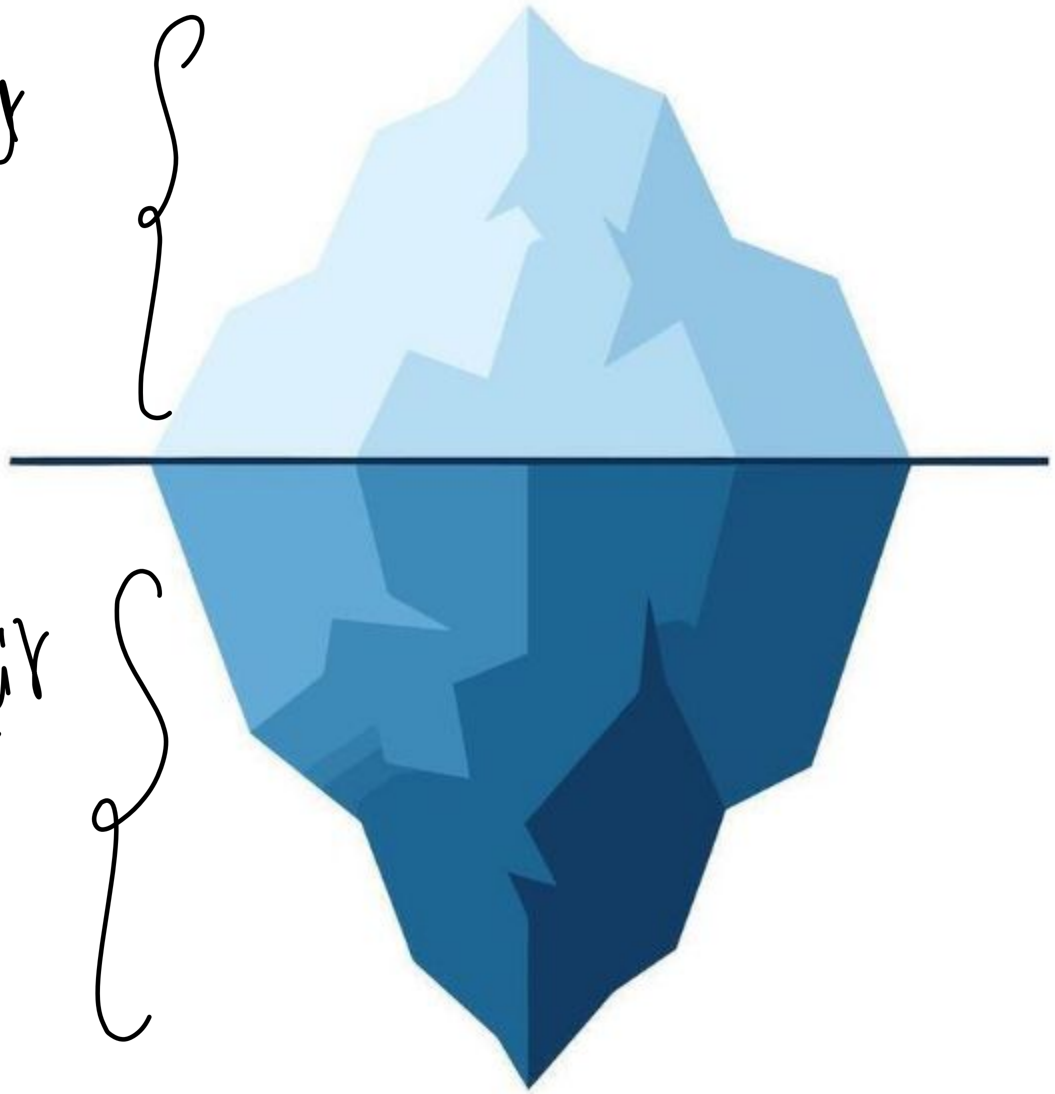
Like, Subscribe, Share, Comment

Implicit (90%)

Watch time, Scroll speed, Rewatch, Time of day, Device, Pause patterns

Explicit

Implicit



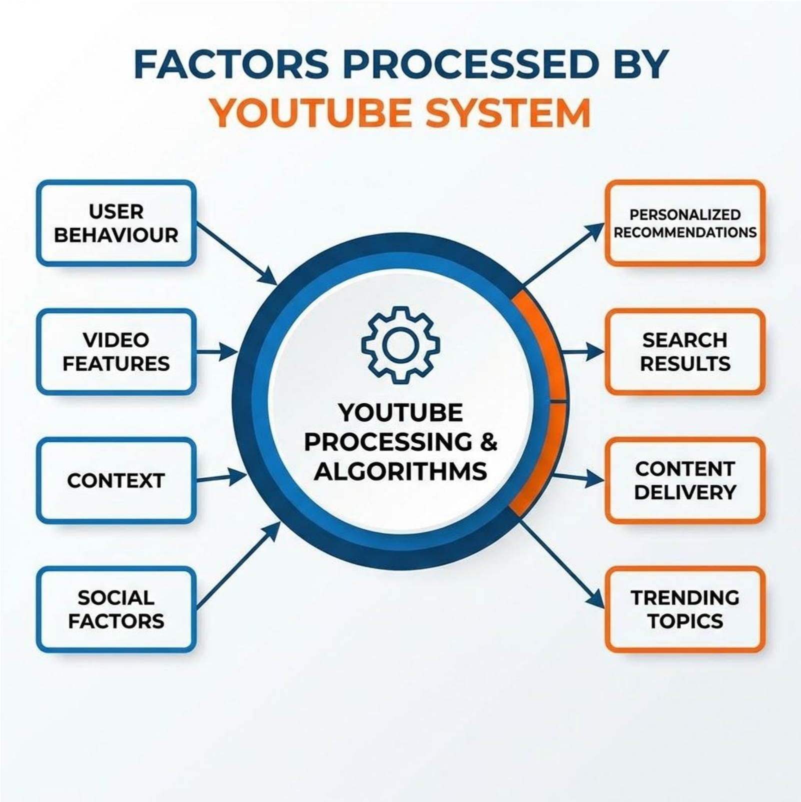
80 BILLION Signals Processed Daily

User Behavior: Watch history, search

Video Features: Title, tags, thumbnail

Context: Time, device, location

Social: Similar users, creator relationships



800M

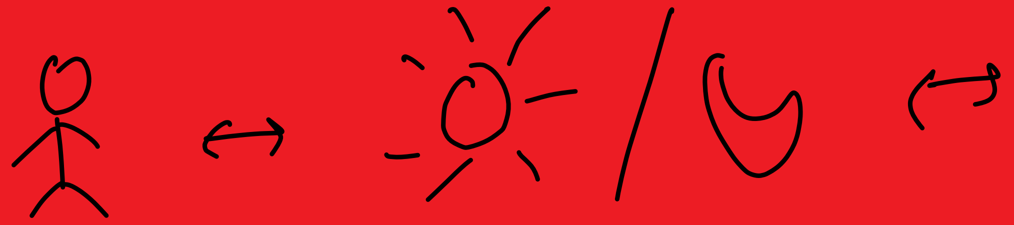
↓

20

~500 ms

The Two-Stage System

How YouTube Narrows 800M to 20



How Do You Find a Needle in a Haystack?

You don't search the whole haystack.

You first remove everything that's obviously NOT a needle.

Think Like a Headhunter

HR Head

Stage 1: Scan

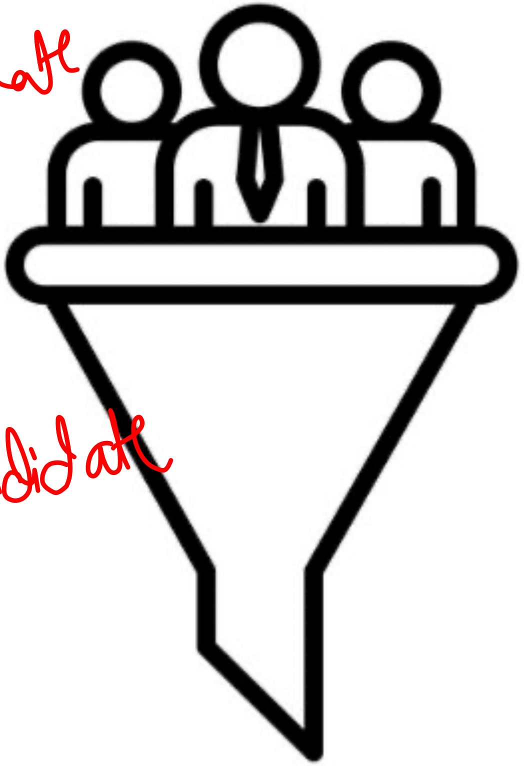
10,000 resumes in 30 seconds each → 500 candidates

Stage 2: Interview

Spend 5 minutes on each of 500 → Top 10 for interview

Best Candidate

Possible Candidate



YouTube's Two-Stage System

800,000,000 Videos

↓

~1,000 Candidates

Stage 1: Candidate Generation

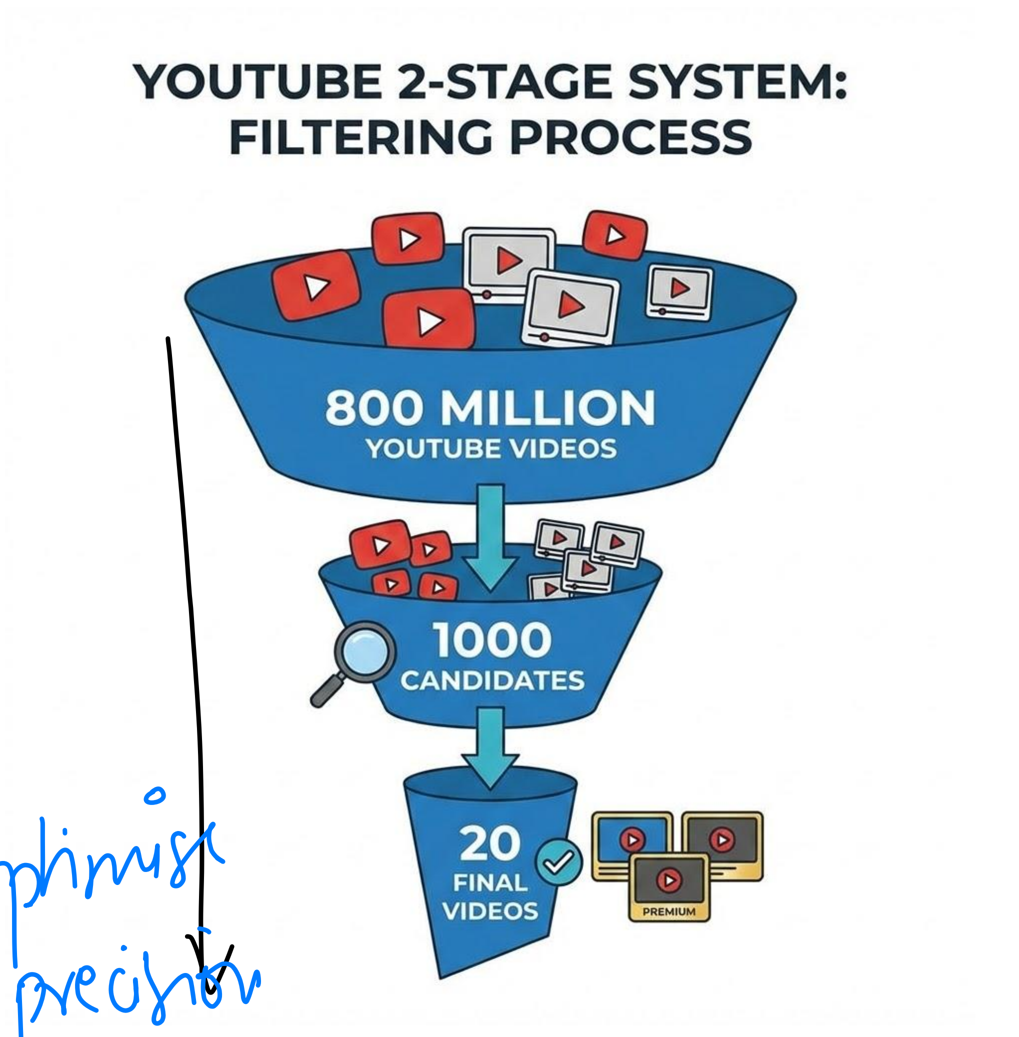
Optimise Recall not precision

↓

20 Videos

Stage 2: Ranking

optimise precision



Stage 1: Candidate Generation

• → Gold

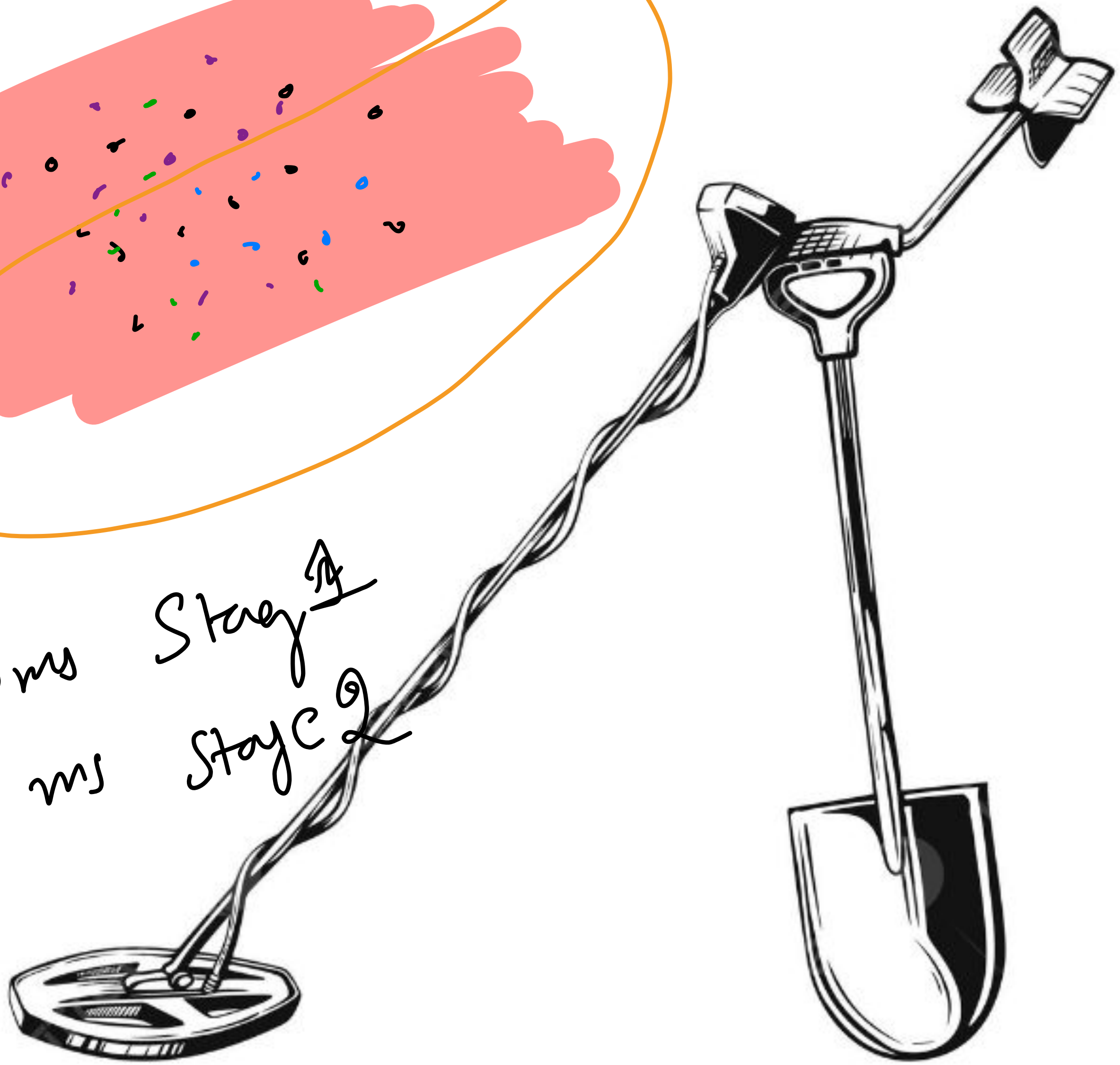
The Fast Filter

- **Goal:** Find ~1,000 "probably relevant" videos
- **Speed:** Milliseconds to process millions
- **Technique:** Embedding retrieval
- **Priority:** Recall over precision

~100ms Stage 1

200 ms Stage 2

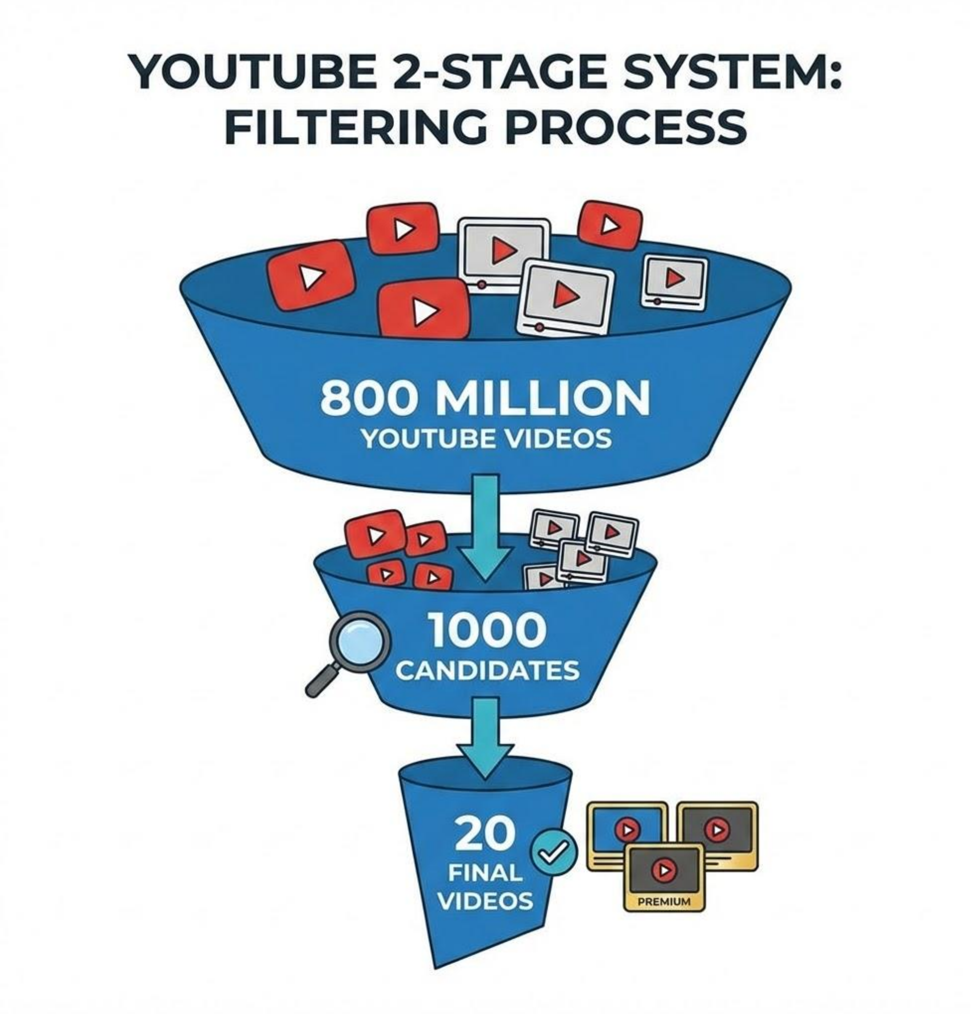
~500 ms



Stage 2: Ranking

The Careful Selection

- **Goal:** Pick BEST 20 from 1,000 candidates
- **Uses:** 100+ features per video
- **Predicts:** Watch time, satisfaction, rewatch likelihood

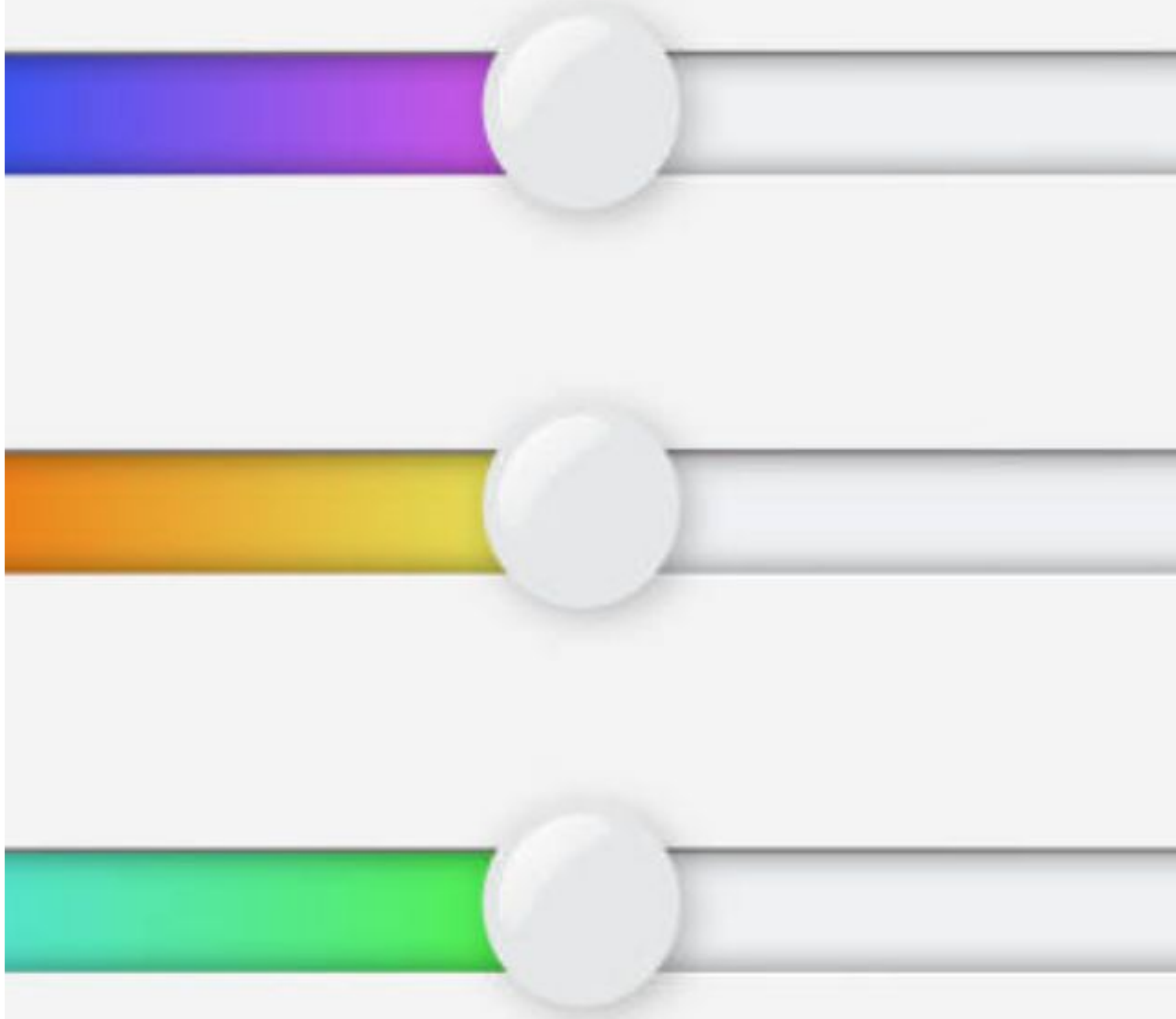


watch time

The Balancing Act

YouTube balances multiple objectives:

- **Watch Time:** "Will they stay engaged?"
- **Satisfaction:** "Likes, surveys, rewatches"
- **Diversity:** "Not all from same topic"
- **Freshness:** "Mix of new & established"
- **Responsibility:** "Avoid harmful content"



Quick Check

In Stage 1, is YouTube optimizing for:

Option A

PRECISION (finding the best)

Option B

RECALL (not missing good ones)

✓ **Correct!**

5

~~10~~ Minute Break

Stretch, refill, reflect

Homework: Look at your YouTube homepage. Why was each video recommended to YOU?

Taste Profiles & Cold Start

You as a Mathematical Point

- → Videos → feature
- → Users → features
- → Videos x Users → features

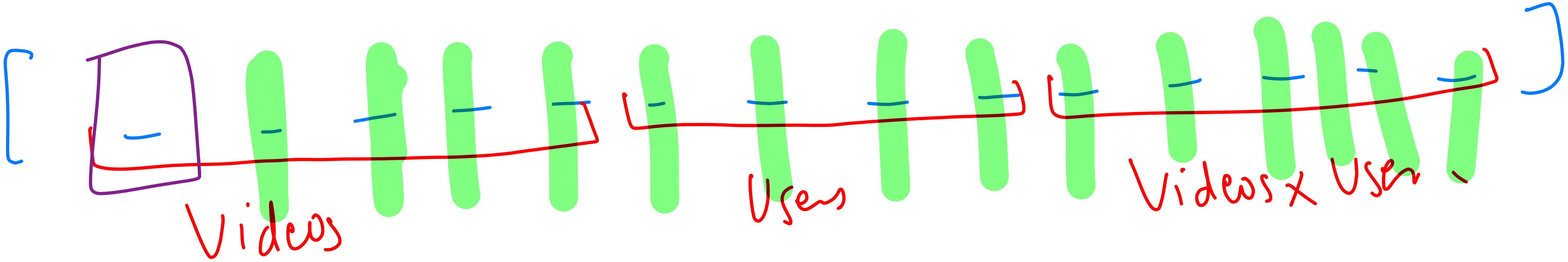
→ Number

→ "

→ "

[- - - - -]

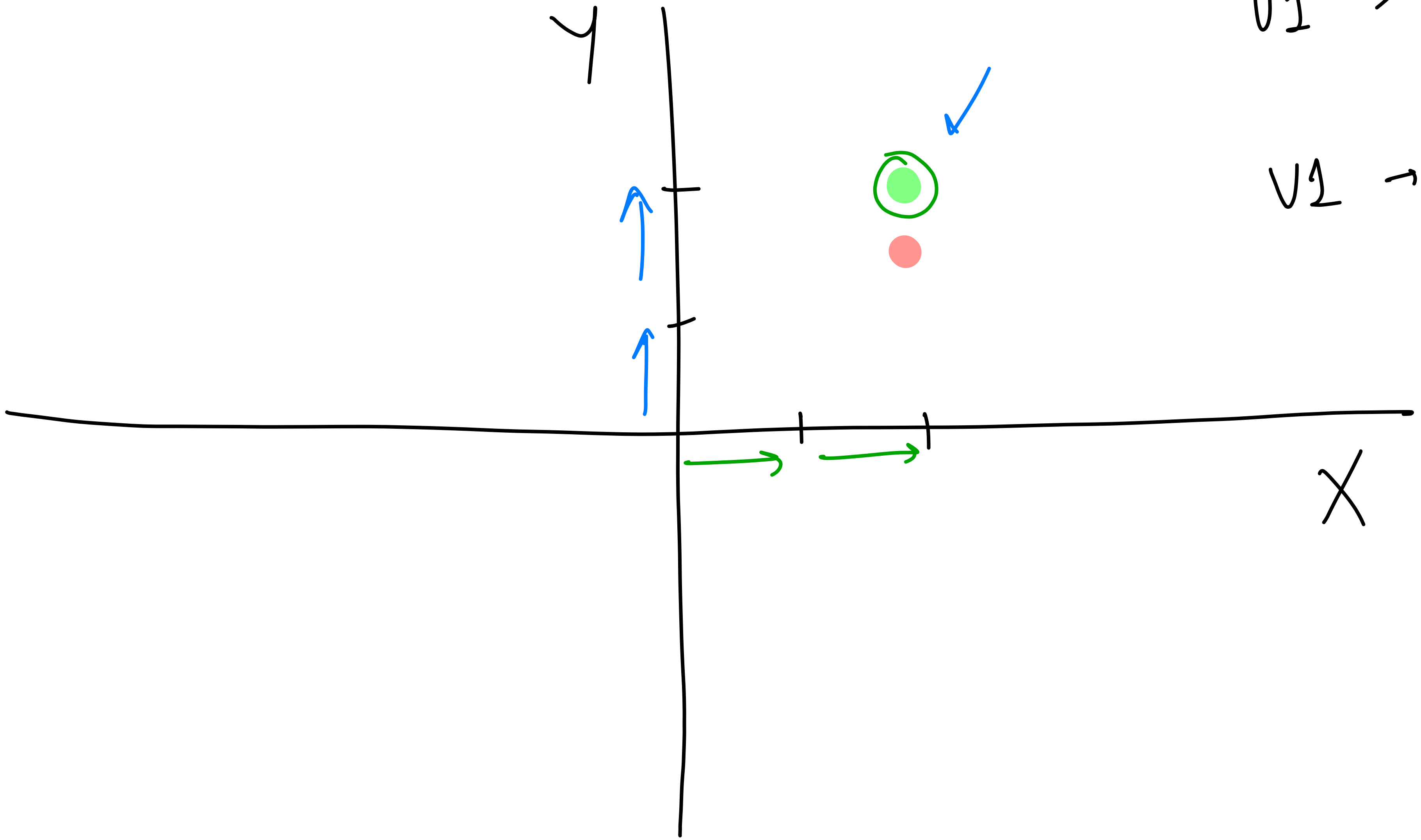
100 +



2.7

$V1 \rightarrow [2,2]$

$V1 \rightarrow [1,2]$

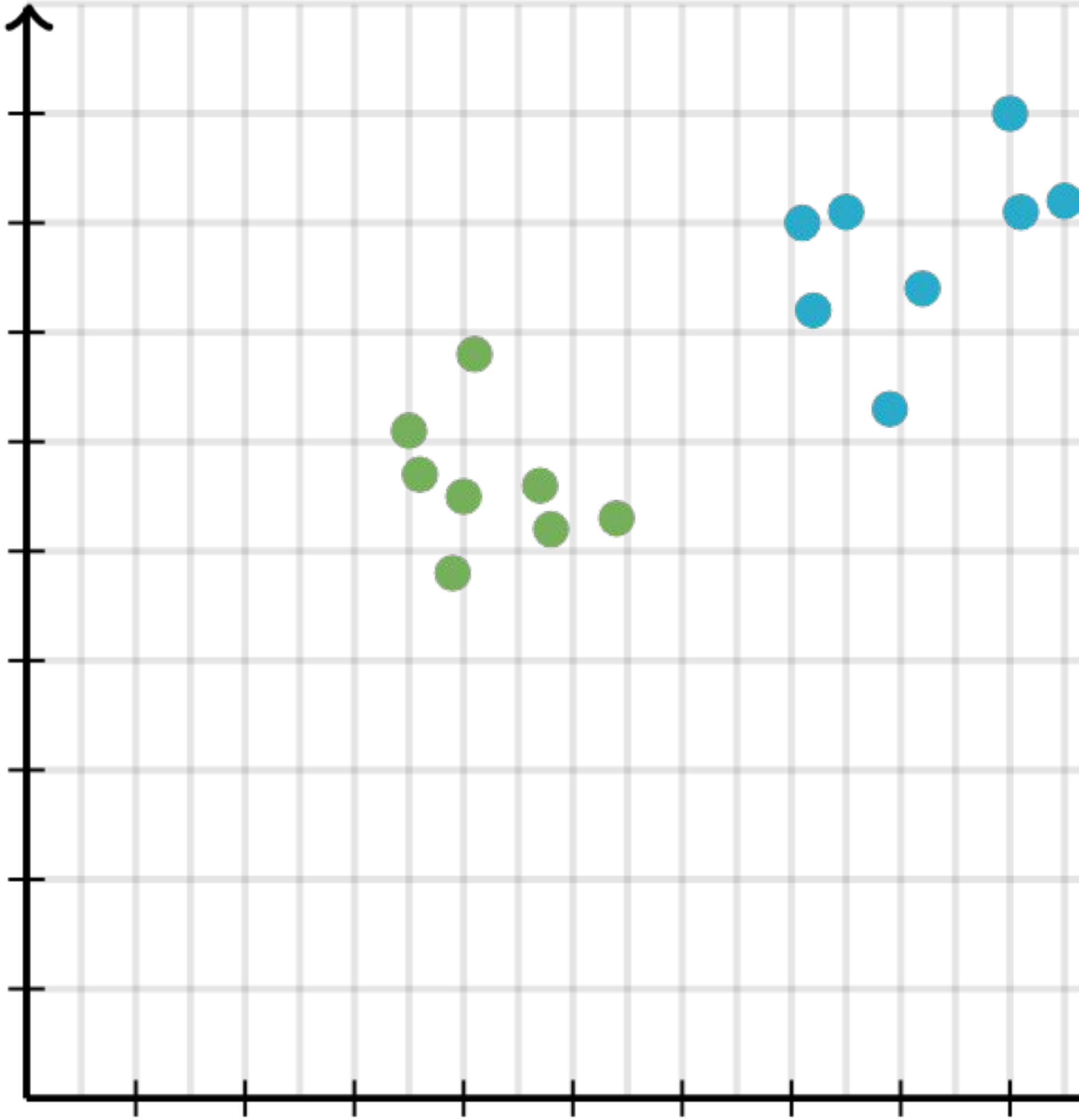


Your Taste = A Point in Space

Imagine a map of interests. Videos and users are both points on this map.

Proximity = Relevance

If your "User Point" is close to a "Video Point", YouTube recommends it.



The Embedding Space (Simplified)

- **X-axis:** Entertainment ← → Education
- **Y-axis:** Short ← → Long

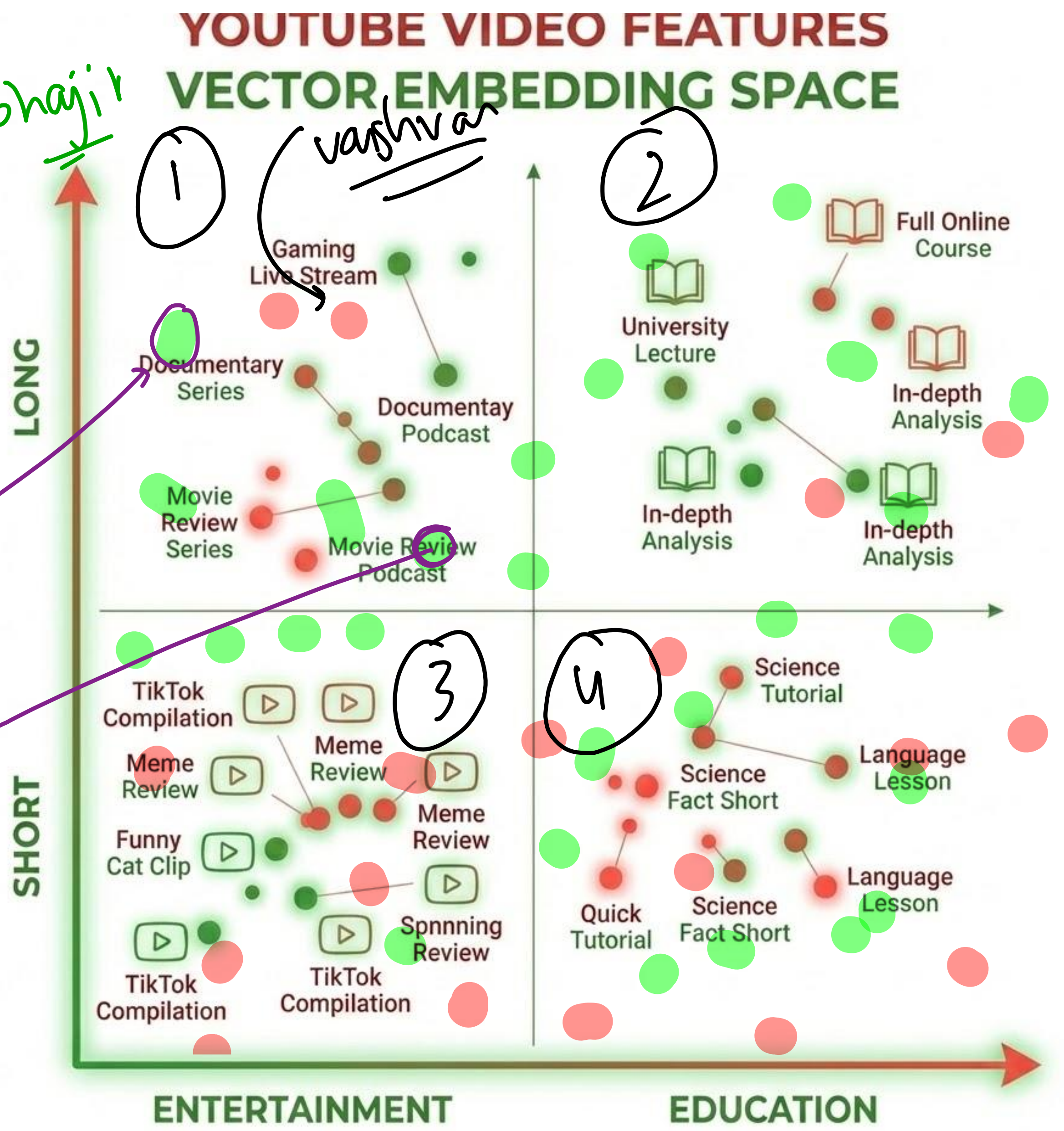
Entertainment
long

Real YouTube uses hundreds of dimensions, not 2. But the concept is identical.

V1

V2

Subhajit



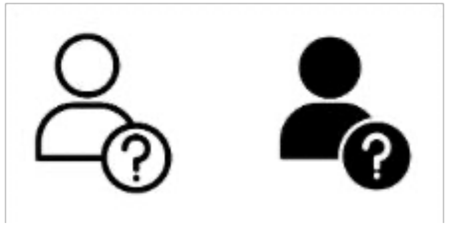
The Cold Start Problem (Again!)

New User

No watch history = Where to place them?

New Video

No engagement = How to rank it?



How YouTube Solves Cold Start

Location

India vs Brazil

Device

iPhone vs Android

Time

7am vs 11pm

Trending

Safe bet for unknowns

Within 10 videos, YouTube has a decent model.

2024 Update: Small Creator Boost

Exploration vs Exploitation

YouTube now shows at least ONE video from smaller/newer creators on every homepage.

YouTube sometimes recommends unknown videos just to LEARN if users might like them.

The Dark Side

Rabbit Holes & Filter Bubbles

The Rabbit Hole Effect

"How did I get HERE?"

> Video A → Video B → Video C → Video D
>
> *You start with 'how to boil an egg.' Two hours later, you're watching a documentary about ancient civilizations.*

The Filter Bubble

- You like X
- YouTube shows more X
- You like X more
- YouTube shows ONLY X

Result: You never see alternative viewpoints.

Does YouTube Radicalize?

- **2020-2022:** Some studies found pathways to extreme content
- **2024-2025:** Algorithm has MODERATING effect
- **Key Finding:** USER CHOICE drives more than algorithm

"If users follow recommendations exclusively, they consume LESS partisan content" — UPenn Study 2024

What YouTube Does About It (2024-2025)

800 m → 20

- 70% reduction in borderline content recommendations
- Information panels for sensitive topics
- "Take a break" reminders
- 192,000+ videos removed in Q1 2025

The Big Question

Should YouTube optimize for:

Option A

What users WANT to watch

Option B

What's GOOD for users

Who decides what's "good"?

Feedback Loops

The System Never Stops Learning

1 Billion Learning Cycles Per Day

1. YouTube recommends
2. You watch (or don't)
3. YouTube learns
4. Profile updates
5. Better recommendations

Every single interaction teaches the algorithm.

What You'll Learn to Build

YouTube Concept	Course Module	When
Embeddings	NumPy + Vector Algebra	Weeks 3-4
Candidate Generation	ML Basics + K-NN	Weeks 9-10
Ranking Models	Regression + Trees	Weeks 13-16
Neural Networks	AIM201	Course 2

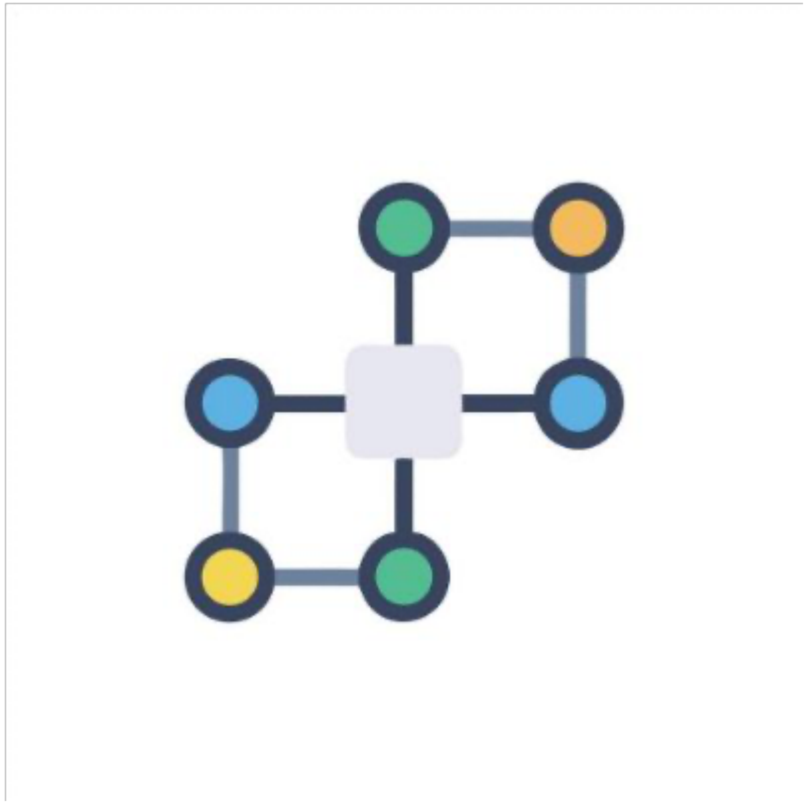
This Pattern Is Everywhere

Netflix

Spotify

Amazon

LinkedIn



Once you understand YouTube's system, you understand MOST recommendation systems.

5 Things to Remember

- YouTube filters 800M → 20 using two stages
- Watch time > clicks. Implicit > explicit.
- Your taste = a mathematical vector
- Filter bubbles exist, but algorithm is moderating
- This pattern powers the entire modern internet

YouTube is Math.

Math you CAN learn.

$$=$$

Math you WILL learn.

Thank You

See you in the next session!